
Snack Choices and Decision Times under the Drift-Diffusion Model*

Tyler Sotomayor[§]

Columbia University

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Abstract

I estimate a drift-diffusion model of snack choices in Experiment 1, using differences in bids to measure relative value. My 435 choices yield a choice-sensitivity estimate of 0.408, compared with 0.308 for the 30,450 choices pooled across the class. The fitted probabilities rise with the bid difference in both samples. Pooled choices change more sharply near equal bids and remain less extreme in the tails than the fitted curve predicts. Response times provide a second check because their shape depends on the same choice parameter. Holding that estimate fixed, I fit the time scale and nondecision time to 15 bin averages. The model explains 55.4 percent of the variation across my trimmed response-time means and 96.1 percent across the class means. The class averages follow the predicted inverted-U closely. My data show a less regular pattern, including faster choices near equal bids than in some neighboring bins.

JEL Codes: D81, D87

Keywords: drift-diffusion model, stochastic choice, response time

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1 Introduction

Choosing between two similarly valued snacks should take longer than choosing between a favorite and a much less valued alternative. The drift-diffusion model (DDM) links this prediction to the probability of choosing each good. When values are close, evidence drifts slowly toward either decision threshold; with a larger value difference, choices become more predictable and quicker on average. I use the formulation presented by Woodford (2026, slides 10–11 and 24–26) in Lecture 5.

The individual file, `indv-Data.csv`, records 435 choices under my UNI, `ts3686`. The pooled file, `popData.csv`, supplies 15 class bins. The first good in each individual record is treated as the left good and the second as the right good. Their mean bids from Section 1 of the experiment give v_L and v_R , so the value difference is $\Delta = v_R - v_L$. There are 368 distinct pairs of goods, some of which I encountered more than once; each choice is a separate trial.

2 Individual Choices

Choice Probabilities (1a). With symmetric thresholds and an unbiased starting point, the model in Woodford (2026, slide 10) gives

$$P_\alpha(\Delta) = \frac{e^{\alpha\Delta}}{e^{\alpha\Delta} + e^{-\alpha\Delta}} = \frac{1}{1 + e^{-2\alpha\Delta}}, \quad \alpha > 0. \quad (1)$$

The parameter α measures how strongly choices respond to the bid difference. Since the two possible responses are exhaustive,

$$\Pr(L \mid \Delta) = 1 - P_\alpha(\Delta) = \frac{e^{-\alpha\Delta}}{e^{\alpha\Delta} + e^{-\alpha\Delta}} = P_\alpha(-\Delta).$$

Equal values imply a one-half probability of either response, and reversing the value difference reverses the choice probabilities.

Likelihood (1b). Let $y_i = 1$ if I choose the right good and $y_i = 0$ otherwise, and define $\sigma_i = 2y_i - 1$. Conditional on the observed bid differences and independent trial disturbances, the probability of

the recorded sequence is

$$L(\alpha) = \prod_{i=1}^{435} P_{\alpha}(\Delta_i)^{y_i} [1 - P_{\alpha}(\Delta_i)]^{1-y_i} = \prod_{i=1}^{435} P_{\alpha}(\sigma_i \Delta_i).$$

Taking logs converts the product into a sum, following the likelihood construction in Souther (2026a, pp. 4–5):

$$LL(\alpha) = \sum_{i=1}^{435} \log P_{\alpha}(\sigma_i \Delta_i) = - \sum_{i=1}^{435} \log(1 + e^{-2\alpha\sigma_i \Delta_i}). \quad (2)$$

Maximum Likelihood Estimate (1c). I maximize equation (2) using a logistic regression of y_i on Δ_i with no intercept. Its slope is 2α , so dividing the estimated coefficient by two gives

$$\hat{\alpha} = 0.408, \quad LL(\hat{\alpha}) = -192.61.$$

The score and curvature provide a check on this numerical estimate:

$$LL'(\alpha) = 2 \sum_i \Delta_i [y_i - P_{\alpha}(\Delta_i)], \quad LL''(\alpha) = -4 \sum_i \Delta_i^2 P_{\alpha}(\Delta_i) [1 - P_{\alpha}(\Delta_i)].$$

At the unrounded estimate, $|LL'(\hat{\alpha})| < 10^{-6}$. The second derivative is negative when at least one bid difference is nonzero, making the positive optimum unique. Under this fit, a one-dollar advantage for the right good raises its choice probability to about 69 percent.

Binned Comparison (1d). Let $M = \max_i |\Delta_i|$, about 4.67 dollars. I divide $[-M, M]$ into 15 equal intervals, each of width $2M/15 \approx 0.622$ dollars. Intervals include their left endpoint and exclude their right endpoint, except that the last also includes M . For the set of trials B_j in bin j , with $N_j = |B_j|$, I calculate

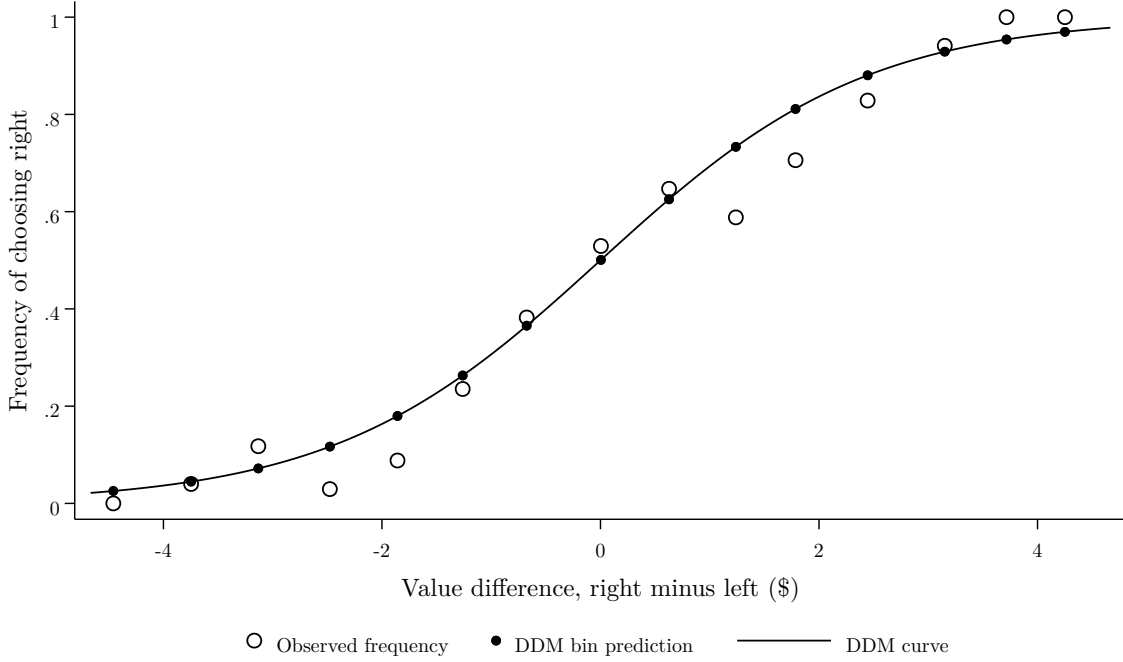
$$p_j = \frac{1}{N_j} \sum_{i \in B_j} y_i, \quad e_j = \frac{1}{N_j} \sum_{i \in B_j} P_{\hat{\alpha}}(\Delta_i).$$

Inverting equation (1) gives the horizontal coordinate,

$$d_j = P_{\hat{\alpha}}^{-1}(e_j) = \frac{1}{2\hat{\alpha}} \log\left(\frac{e_j}{1 - e_j}\right). \quad (3)$$

With this coordinate, $P_{\hat{\alpha}}(d_j) = e_j$, so every predicted bin point lies exactly on the fitted choice curve.

FIGURE 1: INDIVIDUAL CHOICE FREQUENCIES



Notes: Source: `indv-Data.csv`, UNI `ts3686`; 435 trials. Open circles show observed frequencies p_j in 15 equal-width bins. Black dots show mean predicted probabilities e_j at the inverse-probability coordinates in equation (3). The curve is $P_{\hat{\alpha}}(d)$, with $\hat{\alpha} = 0.408$.

I almost always chose the higher-bid snack when the bids were far apart (Figure 1). For nearly equal bids, I chose right in 52.9 percent of trials, close to the predicted 50.1 percent. The intermediate bins are less regular. Near $d_j = 1.24$, the right good won only 20 of 34 choices despite its higher bid, while the model puts its choice probability at 73.3 percent. In the bin near $d_j = -1.86$, I chose right in 8.8 percent of trials, roughly half the predicted 18.0 percent. Across all 15 bins, the root mean squared difference between observed and predicted frequencies, weighted by the number of choices in each bin, is 6.7 percentage points.

3 Individual Response Times

Outlier Removal and Bin Means (2). Across all 435 trials, my mean response time is 1,892 milliseconds and the sample standard deviation is 1,728 milliseconds. Using these values, I exclude

responses at least two standard deviations from the mean and retain those satisfying

$$|RT_i - \overline{RT}| < 2s_{RT}, \quad -1562.61 < RT_i < 5347.44.$$

All 17 excluded responses exceed the upper cutoff. The remaining 418 trials occupy the same 15 bins used for choices, with the original d_j coordinates and choice estimate of α .

The response-time formula in Woodford (2026, slide 10), also developed in Souther (2026b, pp. 3–4), is

$$\mathbb{E}[RT \mid \Delta] = AG_\alpha(\Delta) + t_0, \quad G_\alpha(\Delta) = \begin{cases} \frac{\tanh(\alpha\Delta)}{\Delta}, & \Delta \neq 0, \\ \alpha, & \Delta = 0. \end{cases} \quad (4)$$

Here A sets the time scale and t_0 is time spent outside evidence accumulation. The value at zero follows from $\tanh(x)/x \rightarrow 1$ as $x \rightarrow 0$. This limit matters because one trial has an exactly zero bid difference. The function is even and decreases with $|\Delta|$. For $A > 0$, the model predicts the longest response times when the two values are equal.

Let S_j contain the retained trials in bin j . I average response time and the model’s time function over this same set:

$$T_j = \frac{1}{|S_j|} \sum_{i \in S_j} RT_i, \quad g_j = \frac{1}{|S_j|} \sum_{i \in S_j} G_{\hat{\alpha}}(\Delta_i).$$

I then estimate $T_j = t_0 + Ag_j + u_j$ by ordinary least squares, giving each of the 15 bins equal weight. The estimates are

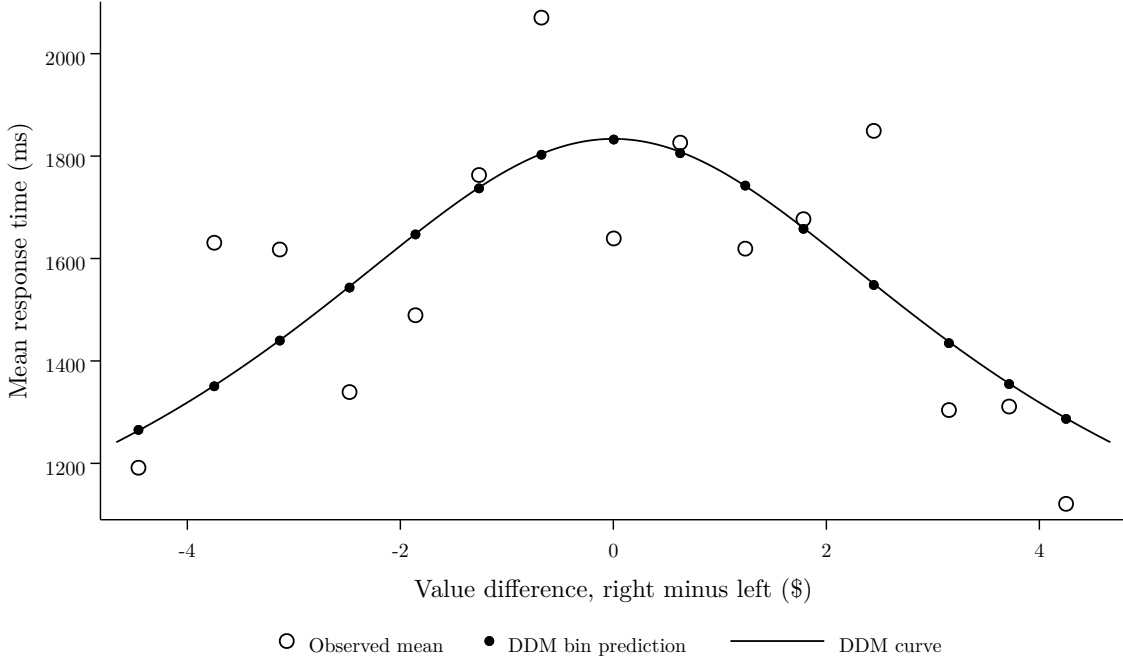
$$\hat{A} = 2913, \quad \hat{t}_0 = 644 \text{ ms}, \quad R^2 = 0.554.$$

Both A and t_0 come out positive, as the model requires.

Assessment. At large bid differences, I usually choose quickly (Figure 2). Near equal bids, the pattern is less regular. The slowest bin, near $d_j = -0.68$, averages 2,070 milliseconds; the next, with nearly equal bids, averages 1,639. I may have found similarly valued snacks easy to choose between, but the pairs near $d_j = -0.68$ harder despite their larger bid gaps.

The bin near $d_j = 2.45$ also stands out. Its mean of 1,849 milliseconds exceeds the fitted value by about 301 milliseconds. Across the 15 means, the model explains 55.4 percent of the variation, with a root mean squared error (RMSE) of 172 milliseconds.

FIGURE 2: INDIVIDUAL MEAN RESPONSE TIMES



Notes: Source: `indv-Data.csv`, UNI `ts3686`; 418 trials within two standard deviations of the full-sample mean. Open circles show T_j ; black dots show $\hat{A}g_j + \hat{t}_0$. The curve is equation (4) evaluated at d . The bin boundaries, d_j , and $\hat{\alpha}$ are held fixed from the choice analysis; both T_j and g_j average over retained trials.

4 Pooled Class Results

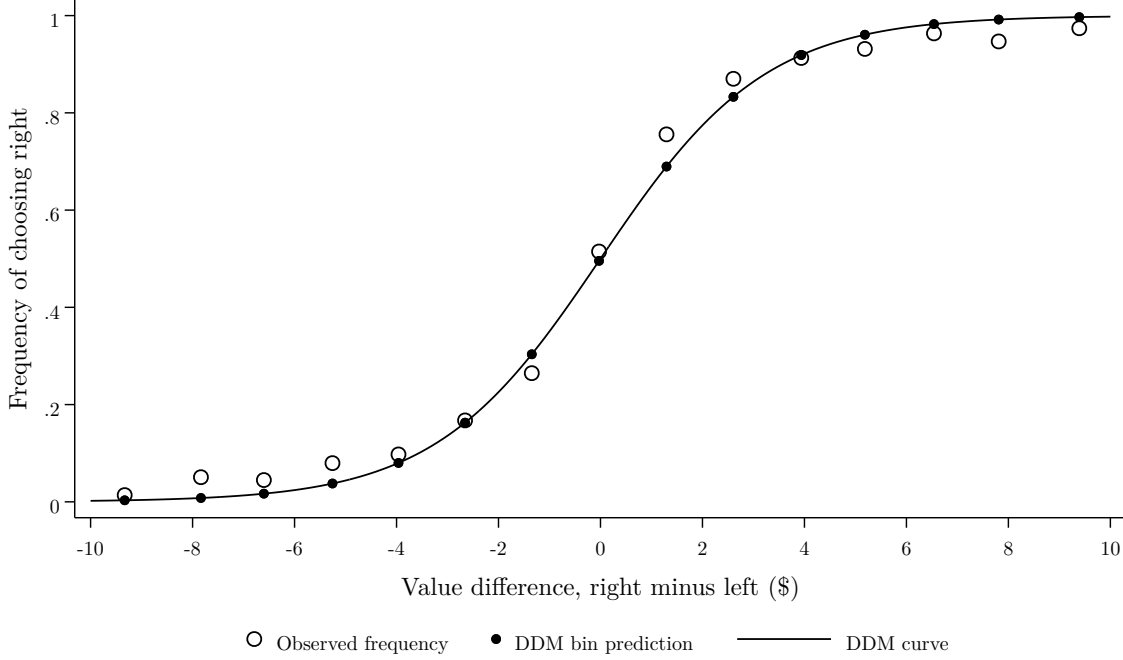
Choice Probabilities (3a). The pooled data contain 30,450 choices, with n_j right choices and m_j left choices in bin j . Treating the supplied mean d_j as each trial's value difference gives the approximate log likelihood

$$LL_c(\alpha) = \sum_{j=1}^{15} [n_j \log P_\alpha(d_j) + m_j \log P_\alpha(-d_j)]. \quad (5)$$

Binomial coefficients can be omitted because they do not depend on α . The subscript c denotes the class data. Maximizing equation (5) gives $\hat{\alpha}_c = 0.308$, with $LL_c(\hat{\alpha}_c) = -14,580.86$.

The plotted frequencies are $p_j = n_j / (n_j + m_j)$ and $e_j = P_{\hat{\alpha}_c}(d_j)$. Under the bin-mean assumption, $P_{\hat{\alpha}_c}^{-1}(e_j) = d_j$ exactly, so the supplied means are also the inverse-probability coordinates used in Question 1(d).

FIGURE 3: POOLED CHOICE FREQUENCIES



Notes: Source: `popData.csv`; 30,450 choices in 15 supplied bins. Open circles show $n_j/(n_j + m_j)$; black dots and the curve show predictions using the class estimate of α . Estimation treats each bin's supplied mean value difference as the value difference on every trial in that bin.

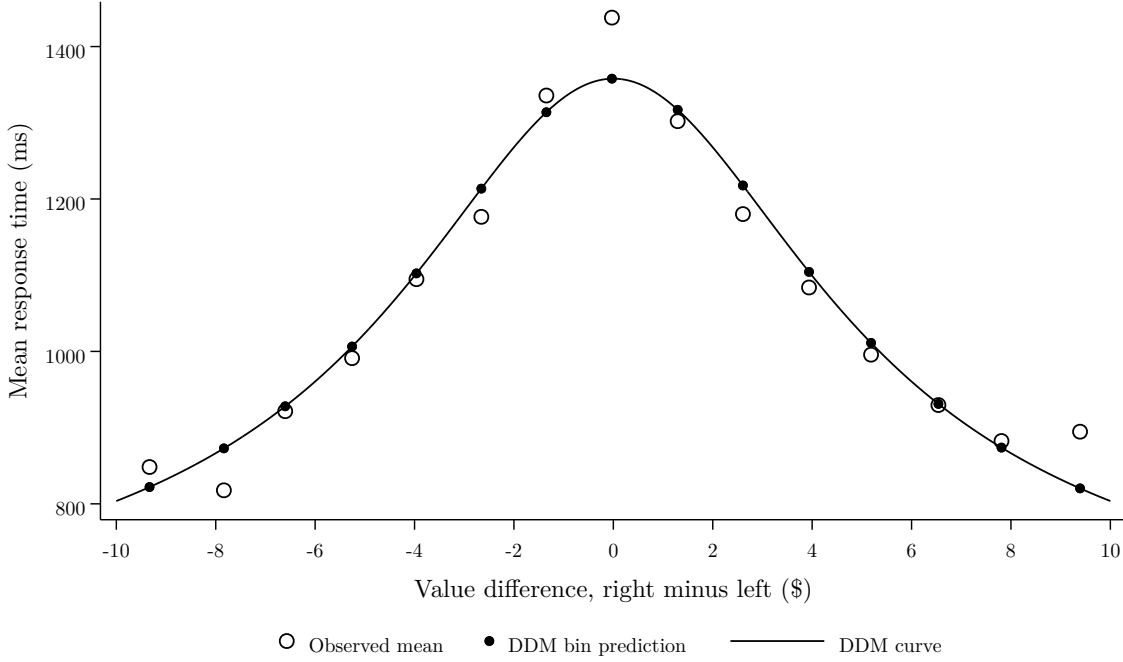
The class choice fit has a count-weighted RMSE of 3.6 percentage points (Figure 3). The errors are systematic. Near $d_j = -1.35$ and $d_j = 1.29$, observed right-choice shares are 26.5 and 75.6 percent; the fitted shares, 30.4 and 69.0 percent, lie closer to one-half. This pattern reverses in the tails. Even with a value advantage of \$7.81, the right snack was passed over in 5.3 percent of choices, against just 0.8 percent predicted. A single α cannot match both the steep change around equal bids and the less extreme choices in the tails.

Mean Response Times (3b). The pooled file contains only bin means, so individual outliers cannot be trimmed. With $\hat{\alpha}_c$ fixed, I regress T_j on $g_j = G_{\hat{\alpha}_c}(d_j)$, giving each of the 15 bins equal weight. The estimates are

$$\hat{A}_c = 2657, \quad \hat{t}_{0,c} = 539 \text{ ms}, \quad R^2 = 0.961.$$

The class means closely follow the predicted inverted-U (Figure 4). Their RMSE is 37 milliseconds, compared with 172 milliseconds for my data; R^2 is 0.961 (Table 1).

FIGURE 4: POOLED MEAN RESPONSE TIMES



Notes: Source: `popData.csv`; 15 supplied response-time means, in milliseconds. Open circles show T_j ; black dots show $\hat{A}_c g_j + \hat{t}_{0,c}$. The time function uses the class estimate of α , and all bins receive equal OLS weight. Under the imposed bin-mean approximation, the black dots lie exactly on the theoretical curve.

The largest residual is in the bin near $d_j = -0.03$, whose mean of 1,438 milliseconds exceeds the fitted value by about 80 milliseconds. At the far right, the class also takes about 74 milliseconds longer than predicted.

TABLE 1: ESTIMATED PARAMETERS AND DESCRIPTIVE FIT

	Individual	Pooled class
Choice trials	435	30,450
Choice sensitivity	0.408351	0.308202
Log likelihood	-192.608	-14,580.856
Choice RMSE (percentage points)	6.70	3.64
Response-time bins	15	15
Time scale, $\hat{A}$	2,913.033	2,656.594
Nondecision time, $\hat{t}_0$ (ms)	644.123	539.137
R^2 of bin means	0.5542	0.9607
Response-time RMSE (ms)	172.21	36.71

Notes: Choice estimates $\hat{\alpha}$ (individual) and $\hat{\alpha}_c$ (class) use maximum likelihood with no intercept. Response-time estimates use unweighted OLS on 15 means, holding the relevant choice estimate fixed. Individual response-time means use 418 retained trials; pooled means are used as supplied. Choice RMSE is $[\sum_j N_j (p_j - e_j)^2 / \sum_j N_j]^{1/2}$, with N_j the total choice count in a bin. Time RMSE divides the residual sum of squares by 15. α is in inverse dollars and A is in milliseconds times dollars.

5 Conclusion

The DDM fits the class's response-time curve more closely than my own. My choices are relatively sensitive to bid differences, yet the bins where I take longest are not those with the closest bids. This matters because the time curve uses the sensitivity estimated from choices; changing A and t_0 cannot move its peak away from equal values. The class means do peak near equal bids.

Both fits depend on how well the bids measure value at the moment of choice. Bids may contain noise or reflect values that changed before the choice task. In the pooled data, averaging across students with different choice sensitivities could also help explain a steeper center and less extreme tails than a single logistic curve predicts. The close fit to class response times does not establish that everyone shares the same α , A , and t_0 .

References

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A Bin Calculations

Tables A1 and A2 report the quantities plotted in the four figures. The Stata and Python code that produces them, together with a standalone replication package, is at github.com/tylersotomayor/ddm-snack-choices.

TABLE A1: INDIVIDUAL BIN CALCULATIONS

Bin	d_j	N_j	N_j^{RT}	p_j	e_j	g_j	T_j	$\hat{T}_j$
1	-4.459	5	4	0.000	0.026	0.2133	1191.5	1265.4
2	-3.747	25	25	0.040	0.045	0.2425	1630.8	1350.6
3	-3.132	34	34	0.118	0.072	0.2731	1617.6	1439.8
4	-2.477	34	33	0.029	0.117	0.3087	1339.3	1543.3
5	-1.858	34	33	0.088	0.180	0.3443	1489.3	1647.1
6	-1.261	34	33	0.235	0.263	0.3752	1763.1	1737.1
7	-0.675	34	33	0.382	0.366	0.3977	2070.4	1802.7
8	0.004	34	31	0.529	0.501	0.4079	1639.2	1832.3
9	0.628	34	31	0.647	0.625	0.3988	1826.3	1805.7
10	1.239	34	30	0.588	0.733	0.3770	1619.3	1742.5
11	1.785	34	34	0.706	0.811	0.3480	1676.9	1657.9
12	2.445	35	34	0.829	0.880	0.3104	1849.3	1548.4
13	3.151	34	33	0.941	0.929	0.2714	1304.1	1434.8
14	3.716	25	25	1.000	0.954	0.2440	1311.0	1354.9
15	4.251	5	5	1.000	0.970	0.2207	1121.0	1286.9

Notes: N_j counts all choices; N_j^{RT} counts retained response-time trials. p_j and e_j use all choices, while T_j and g_j use retained trials. $\hat{T}_j = \hat{A}g_j + \hat{t}_0$. d_j is in dollars and both time columns are in milliseconds. Source: `indv-Data.csv`, UNI `ts3686`.

Averaging Convention. Both g_j and T_j use the 418 retained response-time trials. Averaging $G_{\hat{\alpha}}$ over all original trials instead gives $\hat{A} = 2917$, $\hat{t}_0 = 643$ milliseconds, and $R^2 = 0.555$.

The fitted individual points use the mean of $G_{\hat{\alpha}}$ within each bin, while the smooth curve evaluates it at d_j . The nonlinearity makes the two differ slightly; their maximum vertical separation is 2.7 milliseconds.

TABLE A2: POOLED BIN CALCULATIONS

Bin	d_j	$n_j + m_j$	p_j	e_j	g_j	T_j	$\hat{T}_j$
1	-9.3315	72	0.014	0.003	0.1065	848.3	822.0
2	-7.8358	217	0.051	0.008	0.1256	817.8	872.8
3	-6.6015	424	0.045	0.017	0.1464	921.6	928.0
4	-5.2565	1,193	0.080	0.038	0.1759	991.2	1006.4
5	-3.9613	2,235	0.098	0.080	0.2120	1094.8	1102.4
6	-2.6560	3,552	0.168	0.163	0.2539	1176.6	1213.6
7	-1.3475	4,697	0.265	0.304	0.2916	1335.8	1313.9
8	-0.0281	5,439	0.515	0.496	0.3082	1438.0	1357.9
9	1.2947	4,770	0.756	0.690	0.2928	1302.1	1317.0
10	2.6073	3,649	0.870	0.833	0.2554	1180.2	1217.8
11	3.9377	2,263	0.913	0.919	0.2128	1083.9	1104.3
12	5.1851	1,198	0.932	0.961	0.1777	995.8	1011.2
13	6.5403	438	0.963	0.983	0.1476	929.6	931.2
14	7.8100	226	0.947	0.992	0.1260	882.4	873.8
15	9.3905	77	0.974	0.997	0.1058	894.7	820.3

Notes: The supplied d_j is an arithmetic mean, used as the common value difference for every trial in its bin.

The time function is evaluated at d_j using the class estimate of alpha to give g_j ; $\hat{T}_j = \hat{A}_c g_j + \hat{t}_{0,c}$. d_j is in dollars and both time columns are in milliseconds. Source: `popData.csv`.